

E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 12-A

Roots of functions

(a.k.a. Solutions of nonlinear equations)

Terminology

- An **equation** is an assertion that two expressions (the LHS and the RHS) evaluate to the same number.

e.g. $x^2 - 3x - 4 = 2x + 10$

- To **solve an equation** is to find the values of the unknowns that make it true.
- This is equivalent to finding the **roots (a.k.a. zeros) of a scalar function**.

Informal notation:

$$f(x) = x^2 - 5x - 14$$

Formal notation:

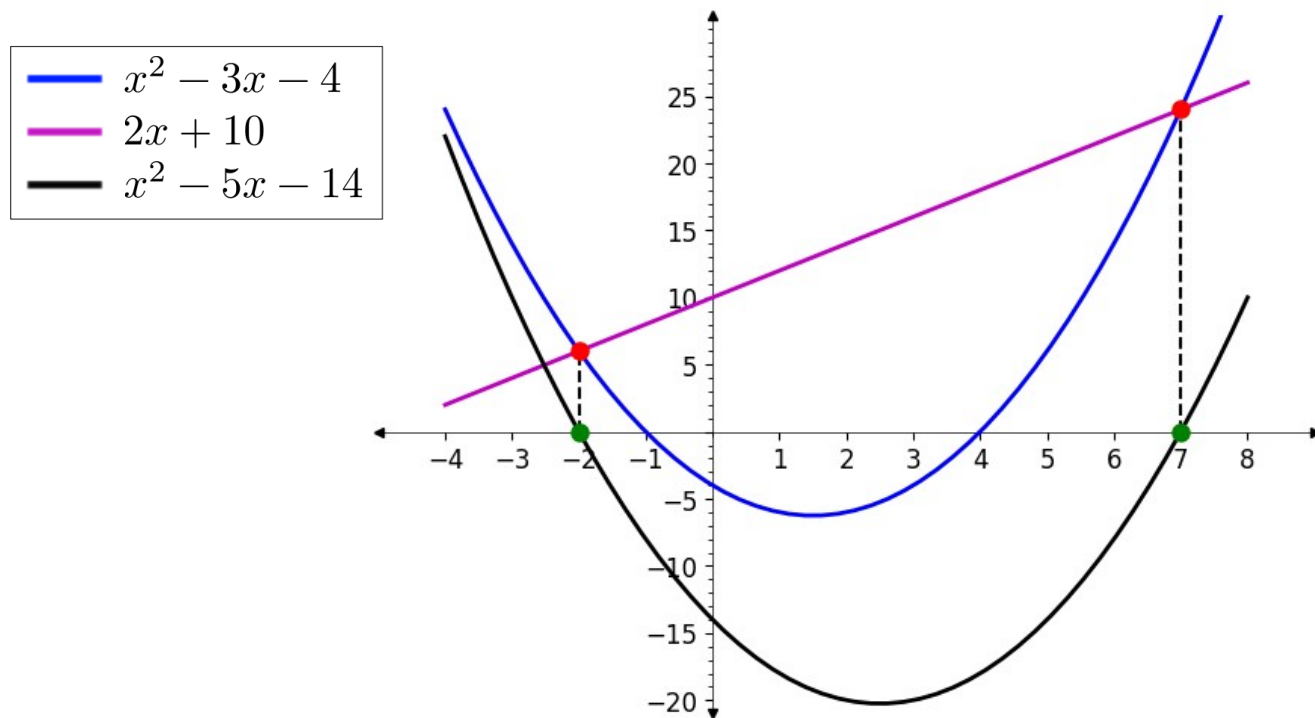
$$f : \mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto x^2 - 5x - 14$$

Solutions of
 $x^2 - 3x - 4 = 2x + 10$

same as

Roots of
 $f(x) = x^2 - 5x - 14$



Terminology system/vector case

- An ***system of equations*** is an assertion that multiple equations hold simultaneously

e.g.
$$\begin{cases} x^2 - y = 1 \\ y^2 - x = 1 \end{cases}$$

- To ***solve a system of equations*** is to find the values of the unknowns that make all of its equations simultaneously true.
- This is equivalent to finding the ***roots (a.k.a. zeros) of a vector function***.

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$(x, y) \mapsto (x^2 - y - 1, y^2 - x - 1)$$

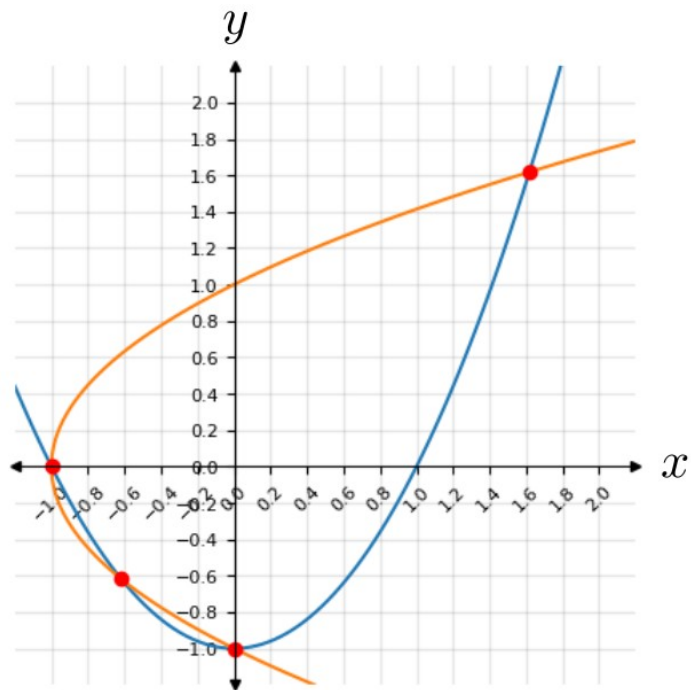
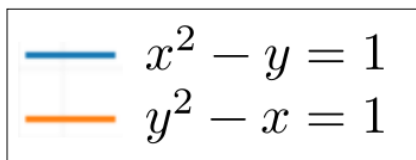
Solutions of

$$\begin{cases} x^2 - y = 1 \\ y^2 - x = 1 \end{cases}$$

same as

Roots of

$$f : (x, y) \mapsto (x^2 - y - 1, y^2 - x - 1)$$



Analytical formulas for the roots of polynomial functions

Quadratic: $f(x) = ax^2 + bx + c$

$$x_1 = \frac{b + \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{b - \sqrt{b^2 - 4ac}}{2a}$$

Cubic: $f(x) = ax^3 + bx^2 + cx + d$

$$\begin{aligned} x_1 &= -\frac{b}{3a} \\ &\quad - \frac{1}{3a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d + \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \\ &\quad - \frac{1}{3a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d - \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \\ x_2 &= -\frac{b}{3a} \\ &\quad + \frac{1 + i\sqrt{3}}{6a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d + \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \\ &\quad + \frac{1 - i\sqrt{3}}{6a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d - \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \\ x_3 &= -\frac{b}{3a} \\ &\quad + \frac{1 - i\sqrt{3}}{6a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d + \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \\ &\quad + \frac{1 + i\sqrt{3}}{6a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d - \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \end{aligned}$$

Quartic: $f(x) = ax^4 + bx^3 + cx^2 + dx + e$

$$\begin{aligned}
\tau_1 &= -\frac{1}{\sqrt{3}} \sqrt{\frac{1}{3} \frac{2^8 (3^8 - 3ac + 12d)}{3 (20^3 - 9abc + 27c^2 + 27a^2d - 72bd + \sqrt{-432^3 - 3ac + 12d^3} + (20^3 - 9abc + 27c^2 + 27a^2d - 72bd)^2)}} \\
&\quad + \frac{1}{\sqrt{3}} \sqrt{\frac{1}{3} \frac{2^8 (3^8 - 3ac + 12d)}{3 (20^3 - 9abc + 27c^2 + 27a^2d - 72bd + \sqrt{-432^3 - 3ac + 12d^3} + (20^3 - 9abc + 27c^2 + 27a^2d - 72bd)^2)}} \\
\tau_2 &= -\frac{1}{\sqrt{3}} \sqrt{\frac{1}{3} \frac{2^8 (3^8 - 3ac + 12d)}{3 (20^3 - 9abc + 27c^2 + 27a^2d - 72bd + \sqrt{-432^3 - 3ac + 12d^3} + (20^3 - 9abc + 27c^2 + 27a^2d - 72bd)^2)}} \\
&\quad + \frac{1}{\sqrt{3}} \sqrt{\frac{1}{3} \frac{2^8 (3^8 - 3ac + 12d)}{3 (20^3 - 9abc + 27c^2 + 27a^2d - 72bd + \sqrt{-432^3 - 3ac + 12d^3} + (20^3 - 9abc + 27c^2 + 27a^2d - 72bd)^2)}} \\
\tau_3 &= -\frac{1}{\sqrt{3}} \sqrt{\frac{1}{3} \frac{2^8 (3^8 - 3ac + 12d)}{3 (20^3 - 9abc + 27c^2 + 27a^2d - 72bd + \sqrt{-432^3 - 3ac + 12d^3} + (20^3 - 9abc + 27c^2 + 27a^2d - 72bd)^2)}} \\
&\quad + \frac{1}{\sqrt{3}} \sqrt{\frac{1}{3} \frac{2^8 (3^8 - 3ac + 12d)}{3 (20^3 - 9abc + 27c^2 + 27a^2d - 72bd + \sqrt{-432^3 - 3ac + 12d^3} + (20^3 - 9abc + 27c^2 + 27a^2d - 72bd)^2)}} \\
\tau_4 &= -\frac{1}{\sqrt{3}} \sqrt{\frac{1}{3} \frac{2^8 (3^8 - 3ac + 12d)}{3 (20^3 - 9abc + 27c^2 + 27a^2d - 72bd + \sqrt{-432^3 - 3ac + 12d^3} + (20^3 - 9abc + 27c^2 + 27a^2d - 72bd)^2)}} \\
&\quad + \frac{1}{\sqrt{3}} \sqrt{\frac{1}{3} \frac{2^8 (3^8 - 3ac + 12d)}{3 (20^3 - 9abc + 27c^2 + 27a^2d - 72bd + \sqrt{-432^3 - 3ac + 12d^3} + (20^3 - 9abc + 27c^2 + 27a^2d - 72bd)^2)}}
\end{aligned}$$

Quintic: $f(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + g$

Abel–Ruffini theorem (1824) : No formula exists!

Numerical methods

We will study three numerical methods (a.k.a. algorithms) for finding the roots of functions.

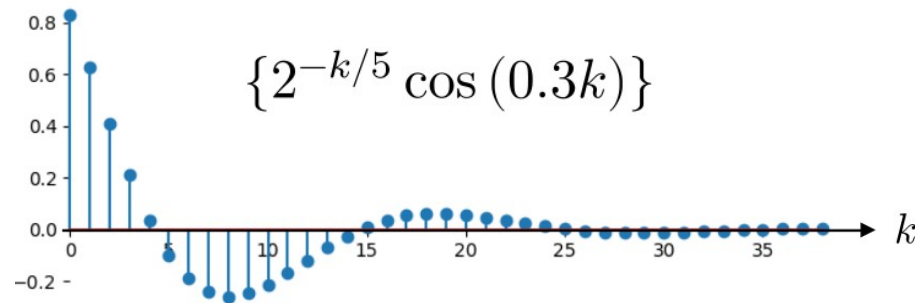
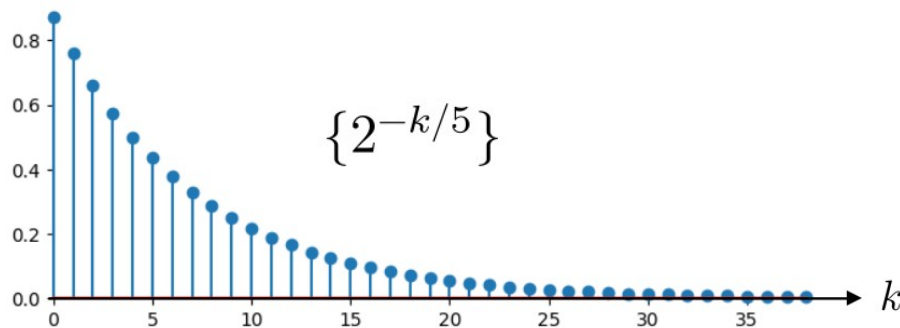
- 1) Bisection
- 2) Newton-Raphson
- 3) Fixed point iteration

Each of these algorithms produces a sequence of numbers that (hopefully) converges to a root.

Sequences and convergence

- A **sequence** is an infinite list of numbers or vectors. $\{x_0, x_1, x_2, \dots\}$
- Notation: $\{x_k\}$
- A **convergent sequence** is a sequence that tends to a fixed value.

Example: Both of these sequences converge to 0.



Bisection

- Requirements: f is $\mathbb{R} \rightarrow \mathbb{R}$ and continuous.

Choose two points N_0 and P_0 such that
 $f(N_0) < 0$ and $f(P_0) > 0$

$k = 0$

repeat until convergence:

$$C = (N_k + P_k)/2$$

if $f(C) < 0$

$$N_{k+1} = C$$

$$P_{k+1} = P_k$$

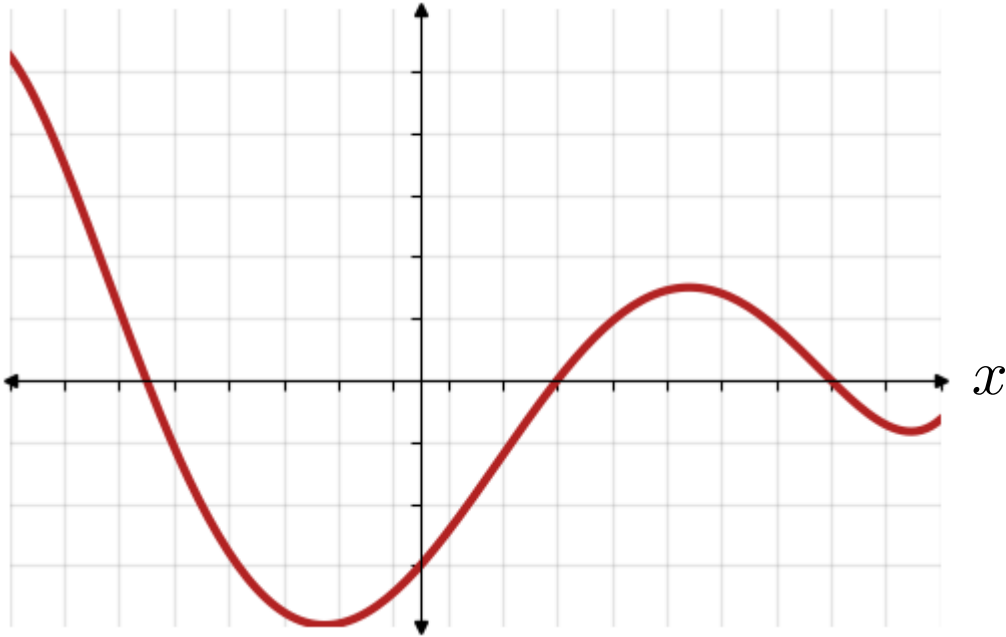
else

$$P_{k+1} = C$$

$$N_{k+1} = N_k$$

$$k = k + 1$$

Example: Bisection



Number of steps to achieve a desired tolerance

- Interval width after k steps

$$|P_k - N_k| = \frac{|P_0 - N_0|}{2^k}$$

- Steps to achieve width smaller than a given ϵ .

Newton-Raphson

- Requirements: f is $\mathbb{R} \rightarrow \mathbb{R}$ and continuous and differentiable.

x_0 is an initial guess

$k = 0$

repeat

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$k = k + 1$

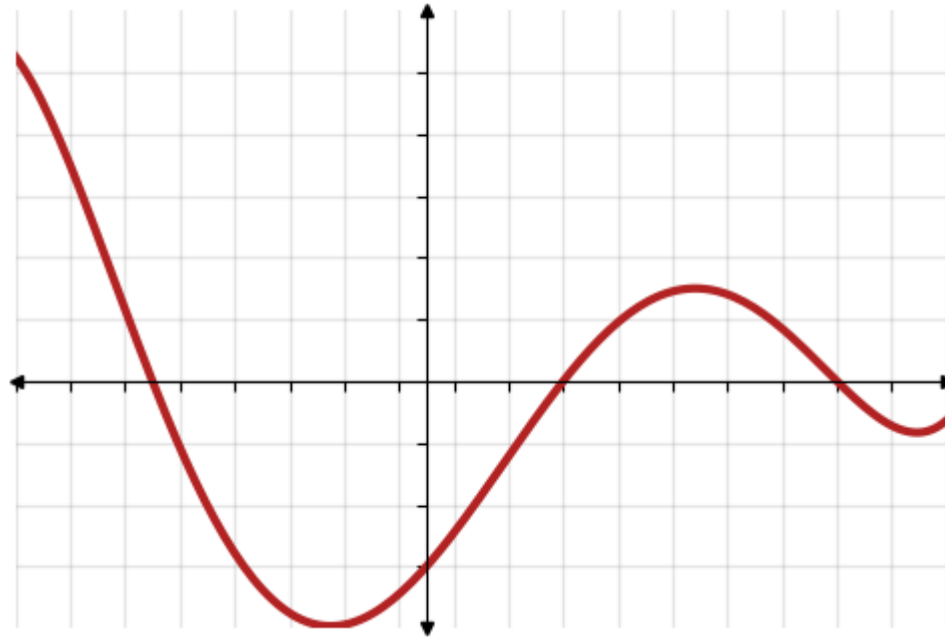
until

$$|x_{k+1} - x_k| < \epsilon_x$$

$$\text{or } |f(x_k)| < \epsilon_f$$

$$\text{or } k > \text{maxsteps}$$

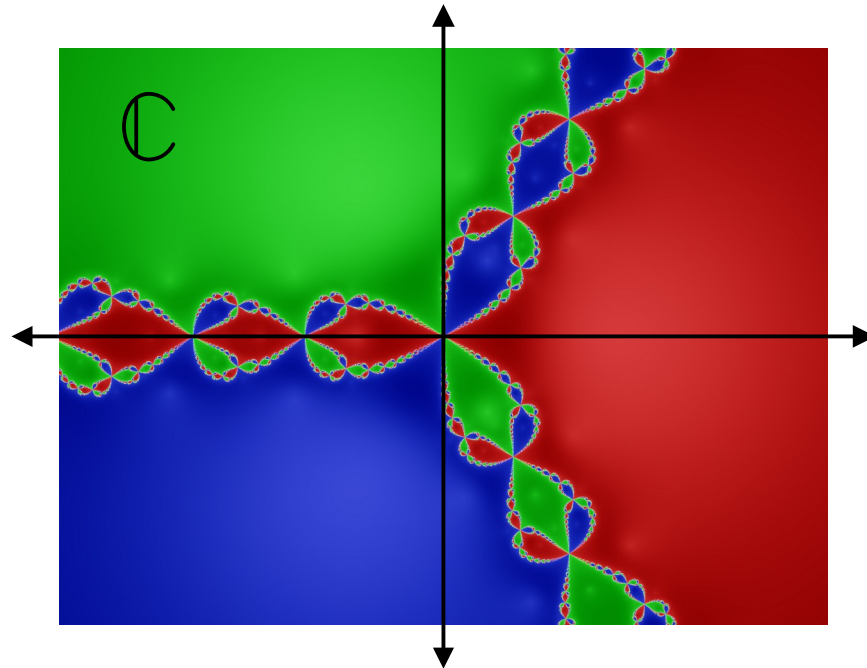
Example: Newton Raphson



Newton-Raphson makes fractals!

$$f(x) : \mathbb{C} \rightarrow \mathbb{C}$$

$$f(x) = x^3 - 1$$



Summary

	Bisection	Newton-Raphson
Requirements	Continuous scalar function.	Continuous function, with derivative.
Initialization	N_0, P_0 with $f(N_0) < 0, f(P_0) > 0$	any x_0
Failure cases	Function has no roots	Function has no roots, Derivative becomes 0